of lipoid materials as contrasted with the ordinary neutral fats. These he thinks have a more specific importance than the general adipose tissue and retain their fat when in hunger the neutral fats waste.

Neutral fat occurs in the blood plasma and in the lymph and chyle, and thus indicates the mode of transport from one part of the body to another, and from the digestive tract to the tissues. It is found in the cells of many parenchymatous organs, such as the liver, adrenals, and others, where it may be merely stored, or where its presence may be explained in other ways which will be discussed later.

There seems to be little evidence to show that such neutral fats, aside from being foods and the source of energy, play any very complex part in the life of the cell. On the contrary, although we do not know exactly how they are distributed in the tissues, we cannot escape the impression that the phosphatides, the cholesterine compounds, and cerebrosides are absolutely essential elements in most of the important functions of the cells. Each new investigation reveals their silent and unsuspected participation in the most fundamental processes.

It is known that even when the microscope, aided by the most effective staining methods, reveals no trace of fat in the tissues, a large quantity can be extracted by chemical methods. A perfectly normal human kidney which shows no recognizable fat yields 10 to 20 per cent. of its weight upon digestion and extraction. This fat must have been in some extremely fine state of division, or else in chemical combination, such that it failed to give the usual staining reactions.

None of the hypotheses about this disposition of lipoids in the cell has up to now been satisfactorily proved, and they are the subject of much dispute. It is Overton's idea that each cell is bounded by a very thin lipoid membrane which controls the entrance and exit of the substances which reach the cell. According to this idea, drugs like narcotics, which are usually soluble in fats, gain easy access to the cell, although it does not appear quite easy to understand how these substances leave the lipoid membrane to enter the watery cell-body. On the other hand, the passage of salts in and out becomes difficult, and the theory is forced to resort to the vital activity of the cell to explain these things. Nevertheless, the idea of a lipoid enveloping membrane for cell and nucleus is very generally held.

Within the cell similar lipoid membranes are supposed to line vacuoles and perhaps to cover some of the specific granules—at any rate, there is much to show that the fine globules of fat which appear in the cell protoplasm do so in relation with mitochondria, or, as Benda claims, rather with the vegetative granules, the plasmosomes. These granules accumulating fat about them finally take on the form of globules. But even with these explanations it is evident that there must be much lipoid material in the cell in an invisible form.

FUNCTIONS OF FATS AND LIPOIDS

The most obvious and best known function of the fats lies in their acting as food-stuffs. In their oxidation to set free energy in the form of heat or work, they require more oxygen than do carbohydrates, so that the respiratory quotient or ratio, $\frac{CO_2}{O}$, is about .795. They form, of course, since they can be conveniently stored, the ideal material for the accumulation of a source of energy. Nevertheless, we must suppose that they also take part in aiding the growth of the tissues through furnishing material for their constructive processes.

If it be true that they form lipoid membranes about each cell, each nucleus, and each vacuole, it must be agreed that they are primarily instrumental in regulating the assimilation of the cell and in permitting it to control in a way the substances which present themselves for absorption.

The part played by lipoid substances, especially the phosphatides and cerebrosides, in the construction of the nervous system, must be of prime importance, although we approach its contemplation so awkwardly by extracting them from the ground-up brain. From their arrangement in the myeline sheaths of the nerve-fibres it would appear that they may act as insulating substances which insure the passage of the nervous impulse to the correct end-organ; in other words, that they serve a purpose analogous to that of the rubber and shellac in a complex electric cable, or even in the brain, to that of the more elaborate insulation in the interior of the dynamo.

In their relation to enzyme action the neutral fats are played upon by lipases which occur everywhere in the organs and fluids of the body, as well as in the digestive juices. Anti-lipases which inhibit this reversible action exist also. The lipases which must exist to control the decomposition of cholesterine esters and of the phosphatides (lecithinase, cholesterase, etc.) are not yet even certainly demonstrated. Bang objects that experiments carried out to show that such lipoid substances may influence the action of other ferments are inconclusive, but Jobling has shown that the decomposition products of some fats—unsaturated fatty acids and their soaps—have the most decisive inhibiting action upon proteolytic ferments, their power being in a sense proportional to the degree of unsaturation of the fatty acid. So universally is it true that such unsaturated fatty acids can impede the action of proteolytic ferments that many pathological conditions (such as the persistence of caseous tuberculous material in its solid form) can be shown to be due to their presence. If they are rendered impotent by saturation of their unsaturated group with iodine, the proteolysis goes on rapidly and the caseous tubercle or gumma rapidly softens.

In the complex process which occurs in the clotting of blood Howell has shown that the thromboplastic substance derived from the tissue is a lipoid, kephaline. It has been shown that certain lipoid substances, especially cholesterine, can act as inhibiting or neutralizing agents toward such hæmolytic poisons as saponin, cobra poison, etc., through forming with them an innocuous compound. Hanes showed that the relative immunity of puppies from chloroform poisoning is due to the large amount of cholesterin esters in their tissues. When artificially introduced into the tissues of adult animals a similar protection is conferred. In the so-called Wassermann reaction lipoid substances, essentially aminomonophosphatides (lecithin, etc.), soluble in alcohol and extracted from heart muscle, are used with the syphilitic serum to combine with the complement and thus withdraw it from the completion of the hæmolysis of sensitized corpuscles, as would occur if the serum were not from a syphilitic. Cholesterol intensifies this action of lecithin.

Our knowledge in this direction is very slight; nevertheless it is enough to suggest the possibility that lipoid substances may sometimes accumulate in an organ for the protection of the cells of that organ from toxic injury. On the other hand, lipoids may act as toxic substances or as activators of toxins. Of these, the toxic ones are foreign lipoids, such as may be extracted from bacteria. The fats from the tubercle bacillus may produce lesions somewhat resembling those caused by the organisms themselves. It is in connection with hæmolytic poisons, such as cobra venom, that lipoids (lecithin) are found to behave as activators. Regarded at first as representing the complement according to Ehrlich's theory (Kyes), it now seems more probable that the lipoids may aid in transferring the poison to the cell, since the "lecithid" is apparently only a solution of the venom in lecithin (Bang). The direct production of immunity against lipoid substances used as antigens has given some vague results, but the matter still remains to be investigated.

Undoubtedly the lipoids fill an important position in many ways, in relation to the processes of immunity, but for the further discussion of the matter reference must be made to works upon that subject (Jobling).

Cholesterine compounds are known to exist in the circulating blood and in the adrenal cortex, as well as in other tissues. What must be a significant index of their importance is found in the course of pregnancy, when there comes a gradual but great increase in the quantity found in the blood, a great storing in the corpora lutea, and, with the end of pregnancy and beginning of lactation, an outpouring of cholesterine esters with the first milk. After that the proportion decreases in the milk, and in the blood sinks back to normal. Why this should be is not known, but the flooding with cholesterine esters seems to have a protective influence of some kind, since under those circumstances animals will survive the loss of the adrenal gland far longer than non-pregnant controls, and, indeed, the injection of cholesterine esters seems to have the same influence (H. A. Stewart).

LITERATURE

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PATHOLOGICAL DISTURBANCES OF FAT

So far we have attempted to review the normal relations of the lipoid substances in the body, and finding our knowledge so woefully incomplete there, we turn not very hopefully to their study under pathological conditions.